La Scuola d'Italia / Dr. Huson / IB Math 12 (a)
28 October 2026

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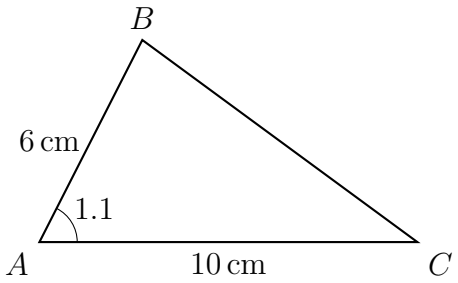
2.6 Quiz: Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \qquad c^2 = a^2 + b^2 - 2ab\cos C \qquad \text{Area} = \frac{1}{2}ab\sin C$$

A calculator is allowed. Give answers correct to three significant figures, unless they are exact. When a question gives an angle in radians, give angles in radians. Diagrams are not to scale.

1. [Maximum mark: 8]

   The following diagram shows triangle ABC, with $AB = 6\,\text{cm}$, $AC = 10\,\text{cm}$ and $B\hat{A}C = 1.1$ radians.



   (a) Find BC. [3]

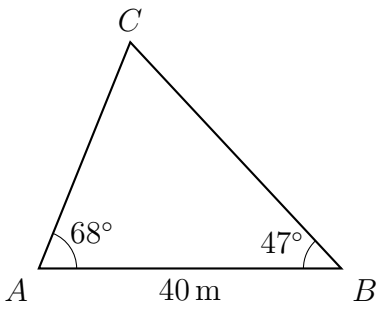
   (b) Find the area of triangle ABC. [2]

   (c) Find $A\hat{C}B$. [3]

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2. [Maximum mark: 9]

A triangular plot of land ABC has $AB = 40\,\text{m}$, $B\hat{A}C = 68°$ and $A\hat{B}C = 47°$.



(a) Write down $A\hat{C}B$. [1]

(b) Find AC. [3]

(c) Find the area of the plot. [2]

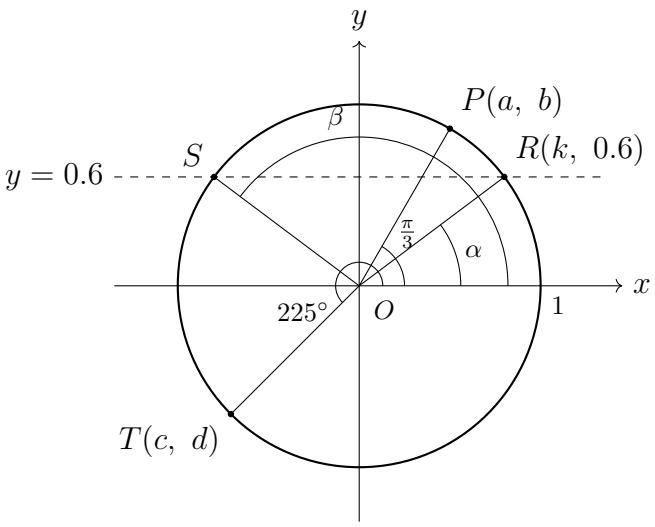
A second triangular plot has two sides of length 24 m and 35 m. Its area is $320\,\text{m}^2$, and the angle between these two sides is obtuse.

(d) Find the angle between these two sides. [3]

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3. [Maximum mark: 10]

The diagram shows the unit circle, centre O. Angles are measured anticlockwise from the positive x-axis. The line $y = 0.6$ meets the circle at R and S.



(a) Write down the exact values of a and b. [2]

(b) Write 225° in radians, as an exact multiple of π. [1]

(c) Write down the exact values of c and d. [2]

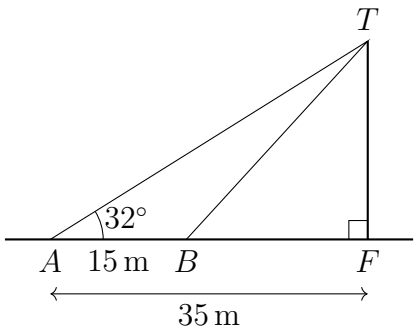
(d) Find the value of k. [2]

(e) The equation $\sin\theta = 0.6$ has two solutions for $0 \leq \theta \leq 2\pi$. They are α and β. Find α and β. [3]

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4. [Maximum mark: 7]

A vertical tree FT has its foot F on horizontal ground. Point A is on the ground, 35 m from F, and the angle of elevation of the top T from A is 32°. Point B lies on $[AF]$, 15 m from A.



(a) Find the height of the tree. [2]

(b) Find the distance AT. [2]

(c) Find the angle of elevation of T from B. [3]

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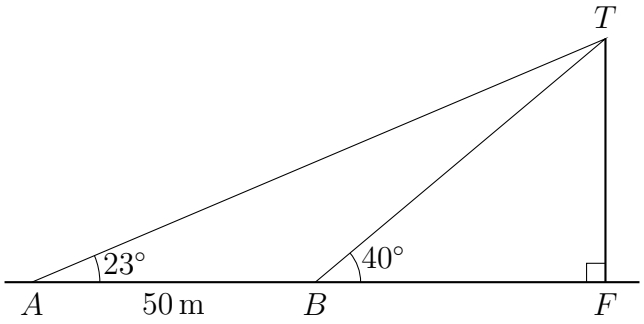
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A calculator is allowed. Give answers correct to three significant figures, unless they are exact. When a question gives an angle in radians, give angles in radians. Diagrams are not to scale.

1. [Maximum mark: 8]

   Points A and B lie on horizontal ground, in a straight line with the foot F of a vertical tower FT, with B between A and F. $AB = 50\,\text{m}$. The angle of elevation of the top T from A is 23°, and from B it is 40°.



   (a) Show that $A\hat{T}B = 17°$. [1]

   (b) Find BT. [3]

   (c) Find the height of the tower. [2]

   (d) Find the distance BF. [2]

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2. [Maximum mark: 8]

In triangle PQR, $PQ = 12\,\text{cm}$, $QR = 9\,\text{cm}$ and $Q\hat{P}R = 0.7$ radians.

(a) Find the two possible values of $P\hat{R}Q$. [3]

In the rest of this question, $P\hat{R}Q$ is obtuse.

(b) Find $P\hat{Q}R$. [1]

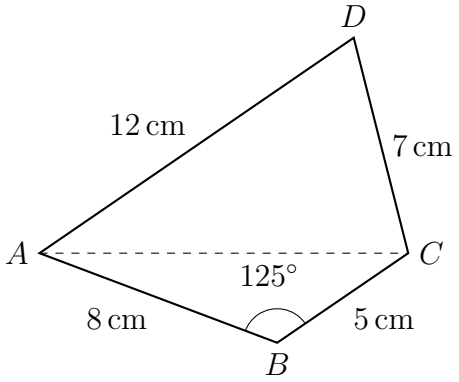
(c) Find PR. [2]

(d) Find the area of triangle PQR. [2]

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3. [Maximum mark: 9]

The following diagram shows quadrilateral $ABCD$, with $AB = 8\,\text{cm}$, $BC = 5\,\text{cm}$, $CD = 7\,\text{cm}$, $AD = 12\,\text{cm}$ and $A\hat{B}C = 125°$.



(a) Find AC. [3]

(b) Find $A\hat{D}C$. [3]

(c) Find the area of quadrilateral $ABCD$. [3]

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4. [Maximum mark: 9]

The table shows the times, t minutes, taken by 40 runners to complete a race.

Time, t	$10 \leq t < 20$	$20 \leq t < 30$	$30 \leq t < 40$	$40 \leq t < 50$	$50 \leq t < 60$
Number of runners	6	p	12	q	3

An estimate of the mean time is 31.25 minutes.

(a) Find the value of p and the value of q. [4]

(b) Find an estimate of the standard deviation of the times. [2]

Every runner is given a time penalty of 2 minutes, and the new times are then written in seconds.

(c) Find an estimate of the mean and of the standard deviation of the new times, in seconds. [3]